Use-Case Flow Specification

UC-02: Reserve Calibrated Equipment

Primary actor: Student **Supporting actor:** University Identity Provider

Preconditions

1. Student has an active university account. 2. Student is authenticated. 3. Equipment and calibration records are available.

Postconditions

Success: Reservation is stored, a unique token is generated, and the slot is blocked from other students.
Failure: No reservation is committed and the slot remains available.

Main Success Scenario

1. Student opens the equipment reservation portal.

2. System displays the equipment catalogue with equipment ID, category, calibration status, and availability.

3. Student selects an equipment item and a preferred start time and end time.

4. System verifies that the student is authenticated and that the requested slot is within the next 7 days.

5. System checks that the equipment is calibrated and that the requested duration does not exceed 2 hours.

6. System checks availability and obtains the required database-level lock for the selected equipment and time interval.

7. System creates the reservation and generates a unique reservation token.

8. System displays the reservation confirmation, including equipment, time slot, and token.

Alternate Flow – Equipment unavailable or not calibrated

5a. If equipment is not calibrated, the system rejects the request and displays that it is unavailable. 5b. If the slot is already reserved, the system informs the student and prompts selection of another time. 5c. No reservation token is generated and no reservation is committed; the student may return to the catalogue.

Business rule: A reservation must not exceed 2 hours and must be made no more than 7 days in advance.